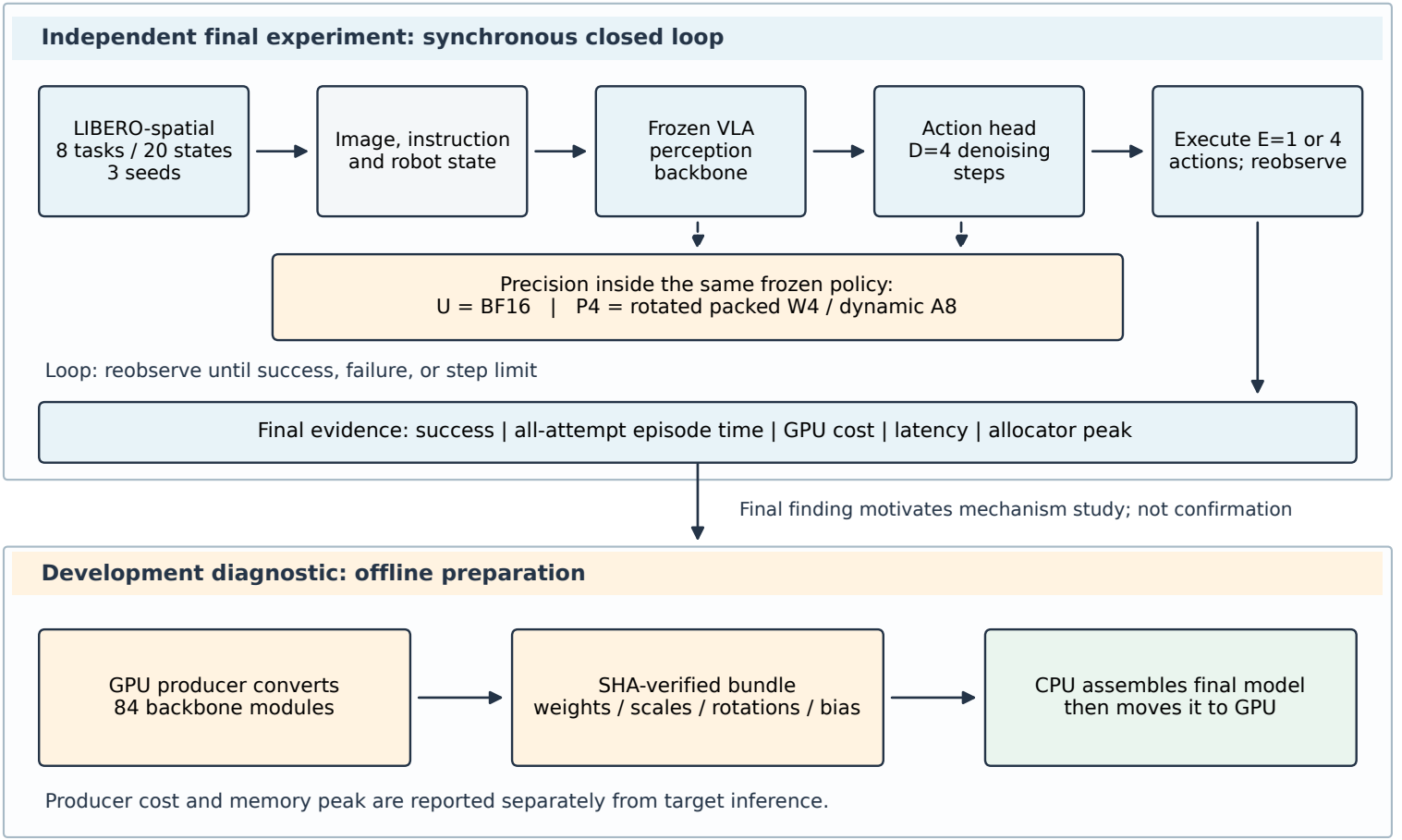
Independent final experiment: synchronous closed loop
LIBERO-spatial
8 tasks / 20 states
3 seeds
Image, instruction
and robot state
Frozen VLA
perception
backbone
Action head
D=4 denoising
steps
Execute E=1 or 4
actions; reobserve
Precision inside the same frozen policy:
U = BF16 | P4 = rotated packed W4 / dynamic A8
Loop: reobserve until success, failure, or step limit
Final evidence: success | all-attempt episode time | GPU cost | latency | allocator peak
Final finding motivates mechanism study; not confirmation
Development diagnostic: offline preparation
GPU producer converts
84 backbone modules
SHA-verified bundle
weights / scales / rotations / bias
CPU assembles final model
then moves it to GPU
Producer cost and memory peak are reported separately from target inference.